STA304 — Term Test 1 Formula Sheet

Estimators, variances, and variance estimates for the population mean, total, and proportion under Simple Random Sampling without replacement (SRS) and Stratified Random Sampling (STRS).

Population Mean

Design	Estimator	Variance	Estimated Variance
SRS	$\bar{y}_{\mathcal{S}} = \frac{1}{n}\sum_{i\in\mathcal{S}} y_i$	$V(\bar{y}_{\mathcal{S}}) = \frac{\sigma^2_{\mathcal{U}}}{n}\left(1-\frac{n}{N}\right)$	$\hat{V}(\bar{y}_{\mathcal{S}}) = \frac{s^2}{n}\left(1-\frac{n}{N}\right)$
STRS	$\bar{y}_{\text{str}} = \sum_{h=1}^{H} \frac{N_h}{N}\,\bar{y}_{h\mathcal{S}}$	$V(\bar{y}_{\text{str}}) = \sum_{h=1}^{H}\left(1-\frac{n_h}{N_h}\right)\left(\frac{N_h}{N}\right)^2 \frac{\sigma^2_{h\mathcal{U}}}{n_h}$	$\hat{V}(\bar{y}_{\text{str}}) = \sum_{h=1}^{H}\left(1-\frac{n_h}{N_h}\right)\left(\frac{N_h}{N}\right)^2 \frac{s^2_h}{n_h}$

$\bar{y}_{h\mathcal{S}}$ = sample mean within stratum h s^2_h = sample variance within stratum h $\sigma^2_{h\mathcal{U}}$ = population variance within stratum h

Population Total

Design	Estimator	Variance	Estimated Variance
SRS	$\hat{t}_{\mathcal{S}} = N\,\bar{y}_{\mathcal{S}}$	$V(\hat{t}_{\mathcal{S}}) = N^2\,\frac{\sigma^2_{\mathcal{U}}}{n}\left(1-\frac{n}{N}\right)$	$\hat{V}(\hat{t}_{\mathcal{S}}) = N^2\,\frac{s^2}{n}\left(1-\frac{n}{N}\right)$
STRS	$\hat{t}_{\text{str}} = \sum_{h=1}^{H} N_h\,\bar{y}_{h\mathcal{S}}$	$V(\hat{t}_{\text{str}}) = \sum_{h=1}^{H}\left(1-\frac{n_h}{N_h}\right) N_h^2\,\frac{\sigma^2_{h\mathcal{U}}}{n_h}$	$\hat{V}(\hat{t}_{\text{str}}) = \sum_{h=1}^{H}\left(1-\frac{n_h}{N_h}\right) N_h^2\,\frac{s^2_h}{n_h}$

$\hat{t}_{\text{str}} = \sum_{h=1}^{H}\hat{t}_h$ where $\hat{t}_h = N_h\,\bar{y}_{h\mathcal{S}}$ is the estimated total within stratum h.

Population Proportion

Design	Estimator	Variance	Estimated Variance
SRS	$\hat{p}_{\mathcal{S}} = \frac{1}{n}\sum_{i\in\mathcal{S}} y_i$	$V(\hat{p}_{\mathcal{S}}) = \frac{p(1-p)}{n}\cdot\frac{N-n}{N-1}$	$\hat{V}(\hat{p}_{\mathcal{S}}) = \frac{\hat{p}_{\mathcal{S}}(1-\hat{p}_{\mathcal{S}})}{n-1}\left(1-\frac{n}{N}\right)$
STRS	$\hat{p}_{\text{str}} = \sum_{h=1}^{H}\frac{N_h}{N}\,\hat{p}_{h\mathcal{S}}$	$V(\hat{p}_{\text{str}}) = \sum_{h=1}^{H}\left(\frac{N_h-n_h}{N_h-1}\right)\left(\frac{N_h}{N}\right)^2\frac{p_h(1-p_h)}{n_h}$	$\hat{V}(\hat{p}_{\text{str}}) = \sum_{h=1}^{H}\left(1-\frac{n_h}{N_h}\right)\left(\frac{N_h}{N}\right)^2\frac{\hat{p}_{h\mathcal{S}}(1-\hat{p}_{h\mathcal{S}})}{n_h-1}$

y_i are binary with values 1 or 0 $\hat{p}_{h\mathcal{S}}$ = sample proportion within stratum h p_h = population proportion within stratum h